

Jing-Zhang Urban AI Testbed: The First Mile from Lab to City

City-in-the-Loop: affected users and operators determine revision, scale, or exit

Rail is not one line: three systems, three operating conditions

Underground HSR and visible Line 13 coexist in the middle; HSR returns to grade in the north

FORMER SURFACE JING-ZHANG / NON-OPERATING HERITAGE

Surface remnants, removed sections, and public-space opportunities

JING-ZHANG HSR / OPERATING

Beijing North at grade → Qinghuayuan Tunnel → returns to grade in the north

METRO LINE 13 / OPERATING

Viaduct, embankment, and surface; visible does not mean crossable

Strategic diagnosis: concentrated innovation, uneven everyday support

Public fact, mapped orientation, provisional extent, and unknowns remain separate

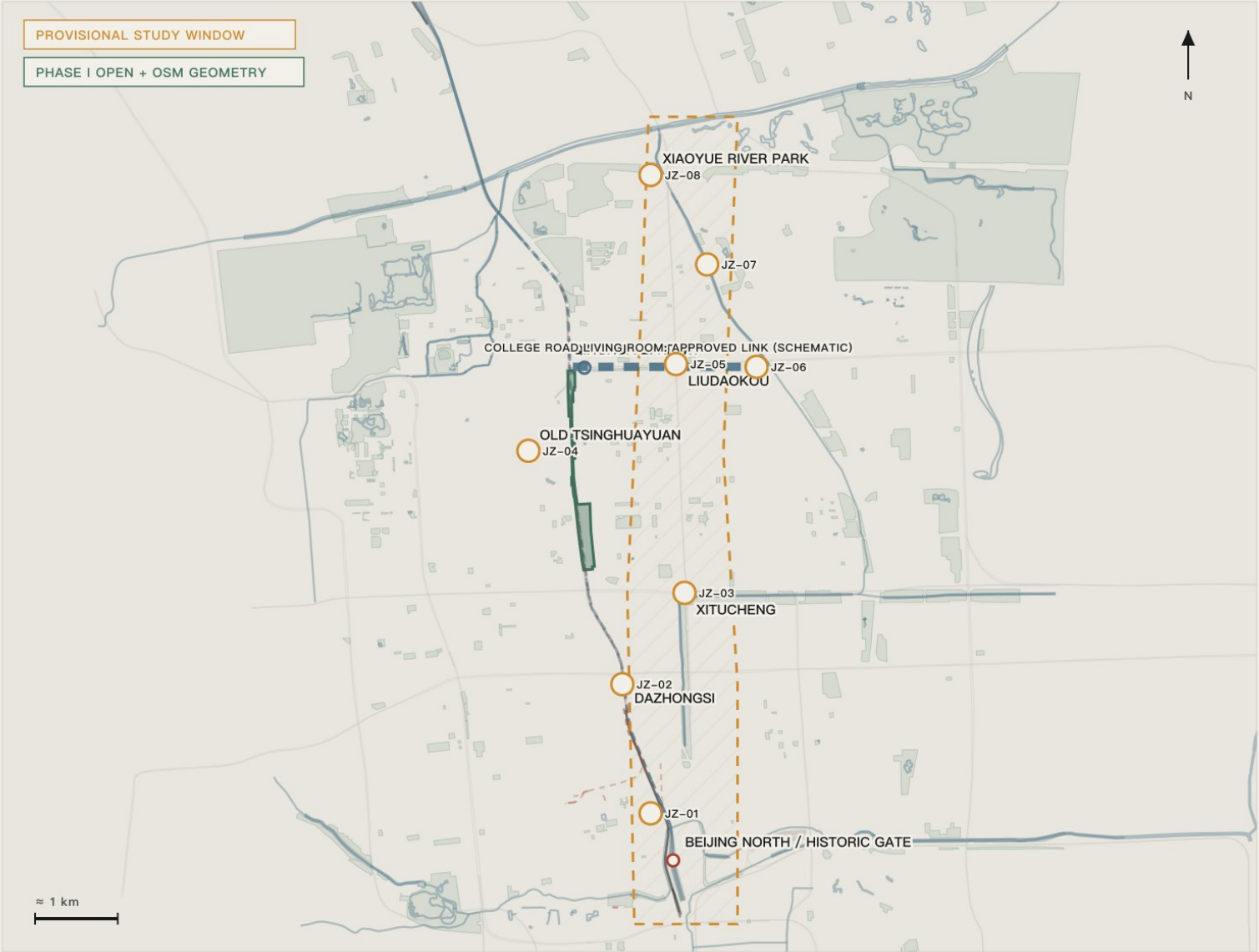
JZ-RCT / 01 / V0.1

01 / HISTORY × REAL SPACE

EVIDENCE LAYERED

WHY JING-ZHANG? AI'S FIRST MILE INTO REAL CITY LIFE

In 1909, Jing-Zhang proved autonomous engineering through terrain and long operation.
Today: can a city understand, take over, correct and exit AI?



Status: green = open; blue rail = active HSR (dashed below grade); black = Line 13; red dash = old rail; amber = proposal.
Sources: Beijing planning, parks and reform authorities; Haidian parks authority; OpenStreetMap contributors, 2026–08–10.

FOUR VERIFIED FACTS

1905—1909

Autonomous survey, design, construction and operation became a working railway

≈ 9 km

Official planning context, Beijing North to the North Fifth Ring; not fully open today

≈ 6.02 km

Active Qinghuayuan Tunnel; HSR remains exposed at its south and north ends

2.5 km / 16.8 ha

Phase I, Qinghua East Road to Zhichun Road, officially open

THEREFORE: NOT AN AI DISPLAY BACKDROP

The heritage spine represents long operation and public space;
cross-links expose access, maintenance and service responsibility;
the real city can stop a trial and return to ordinary use.

Spatial structure: one spine, Three Zones, Two Wings, three testbeds

Repair the ordinary city first; trial only inside bounded, recoverable units

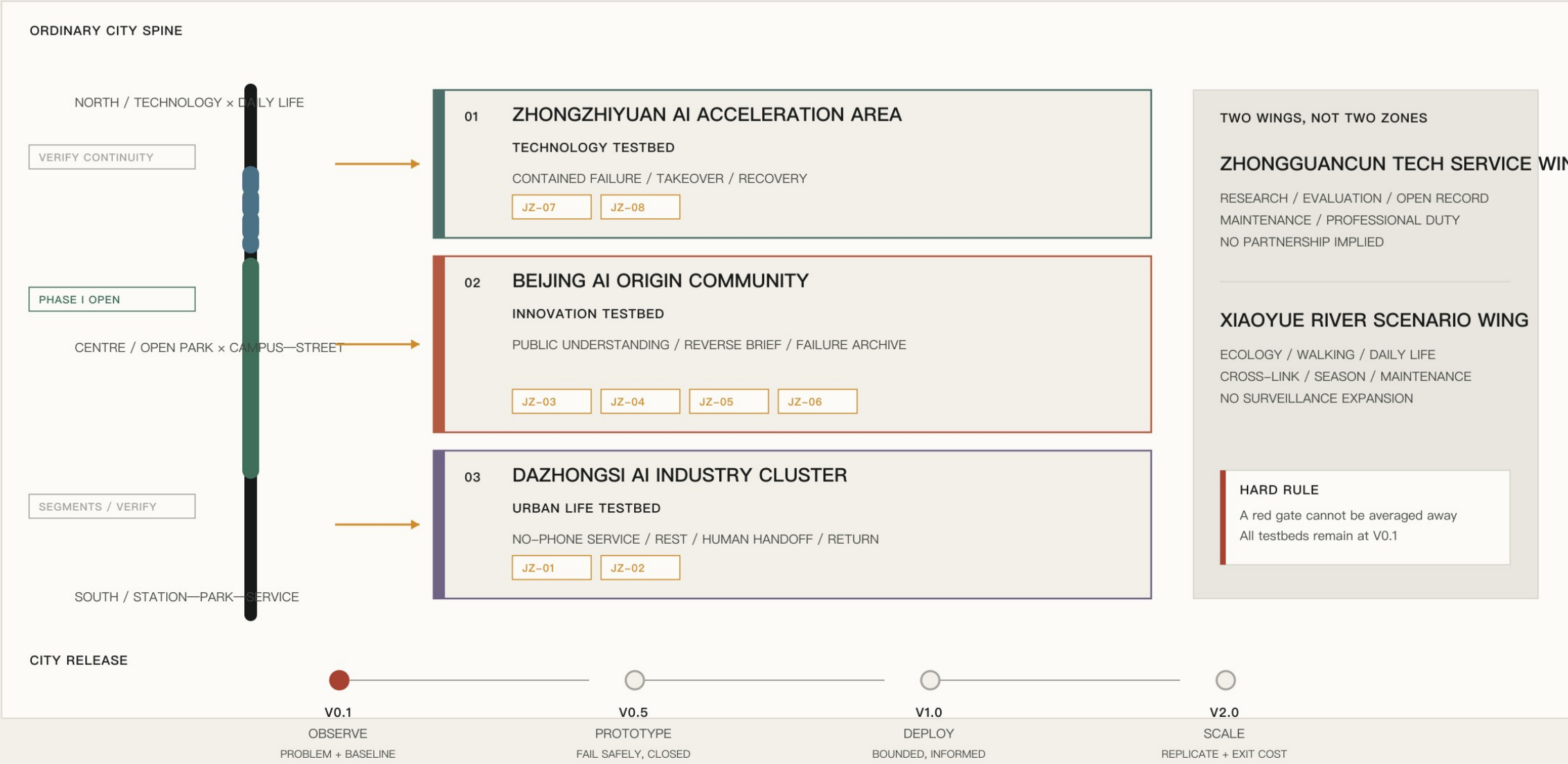
JZ-RCT / 02 / V0.1

02 / SPATIAL STRUCTURE: LAB TO CITY

CONCEPTUAL / NOT ZONING

ONE ORDINARY PUBLIC SPINE
EIGHT CROSS-LINKS, THREE TESTBEDS

This is not a statutory land-use plan.
It shows one causal rule:
repair the ordinary city before a bounded, reversible AI trial.



Structure: organiser-provided three areas/two wings + eight proposal interfaces + City Release. Testbeds are conceptual, not zoning, official boundaries or permits.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Three key areas: ordinary routes, trial boundaries, and hard gates

Study windows are not official polygons; project points are searches and audits, not approvals

JZ-RCT / 03 / V0.1

03 / THREE KEY AREAS: ALIGN FUNCTION, DO NOT INVENT BOUNDARIES

PROVISIONAL / NO OFFICIAL BOUNDARY

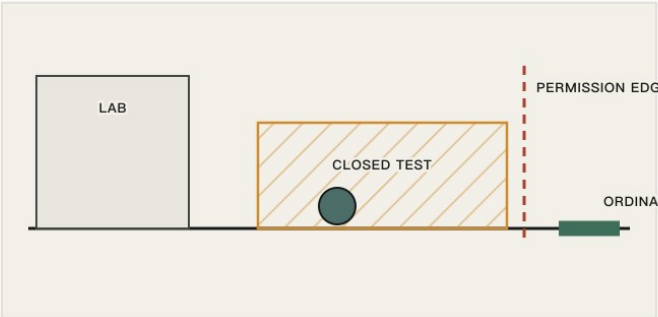
THREE TESTBEDS, NOT SHOWROOMS THREE DISTINCT FAILURE TESTS

Official names and approximate areas remain unchanged.
Without exact boundaries, this figure shows
interfaces, questions and hard gates only.

01192.1 ha

ZHONGZHIYUAN AI ACCELERATION AREA

TECHNOLOGY TESTBED



01 / ORDINARY CITY FIRST

ENCLOSURE / ORDINARY BYPASS / STAFF / FIRE / RECOVERY FIRST

02 / BOUNDED TRIAL

LOW-SPEED MOBILITY, TAKEOVER AND FAILURE START CLOSED

03 / HARD GATE

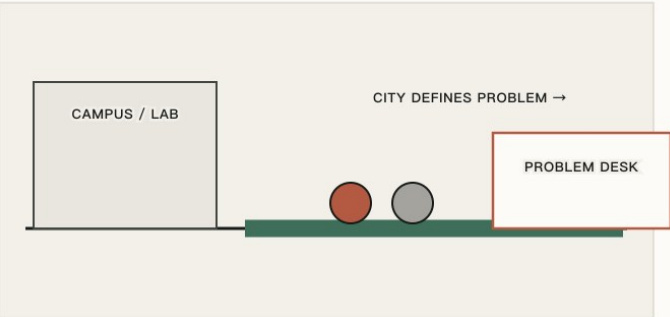
NO WRITTEN SITE, SAFETY OWNER OR INSURANCE → NO TRIAL

NOW: V0.1 / NOT STARTED

02104.3 ha

BEIJING AI ORIGIN COMMUNITY

INNOVATION TESTBED



01 / ORDINARY CITY FIRST

OPEN PARK / SEATING / STATIC EXPLANATION / HUMAN DESK FIRST

02 / BOUNDED TRIAL

PUBLIC PROBLEMS ENTER LABS; FAILURE AND REVISION STAY PUBLIC

03 / HARD GATE

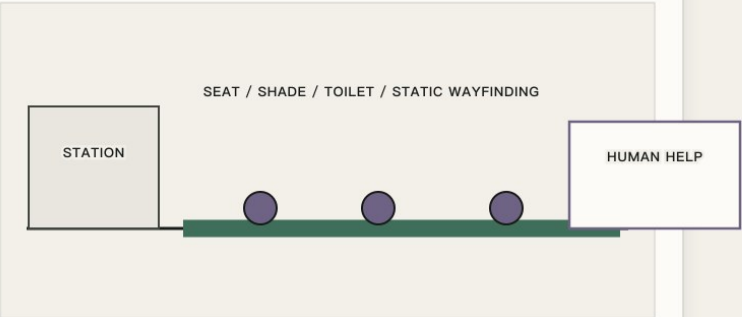
PROBLEM OWNER REJECTS REWRITE → RETURN TO V0.1

NOW: V0.1 / NOT STARTED

0372.0 ha

DAZHONGSI AI INDUSTRY CLUSTER

URBAN LIFE TESTBED



01 / ORDINARY CITY FIRST

NO-PHONE WAYFINDING / REST / TOILET / HUMAN WINDOW FIRST

02 / BOUNDED TRIAL

TEST NON-DIAGNOSTIC JOURNEYS; NO HEALTH SCORE OR TRACKING

03 / HARD GATE

PROV-KEY-003 DRIFTS ≈2.26 KM → DO NOT INFER FROM ITS POLYGON

NOW: V0.1 / NOT STARTED

Approximate organiser–provided areas: 192.1 / 104.3 / 72.0 ha. Diagrams are interface concepts, not parcel plans, official boundaries, construction or partnerships.

Zhongzhiyuan: Technology Testbed

Keep controlled trials inside; repair ordinary movement gaps first

03--N

众智园 | 受控试验留在院内，普通通行优先补断点

192.1 ha detailed boundary pending / mapped window 1.8 x 1.4 km / both maps 1:10000

A3 REVIEW V1

BOUNDARY PENDING / 不提交

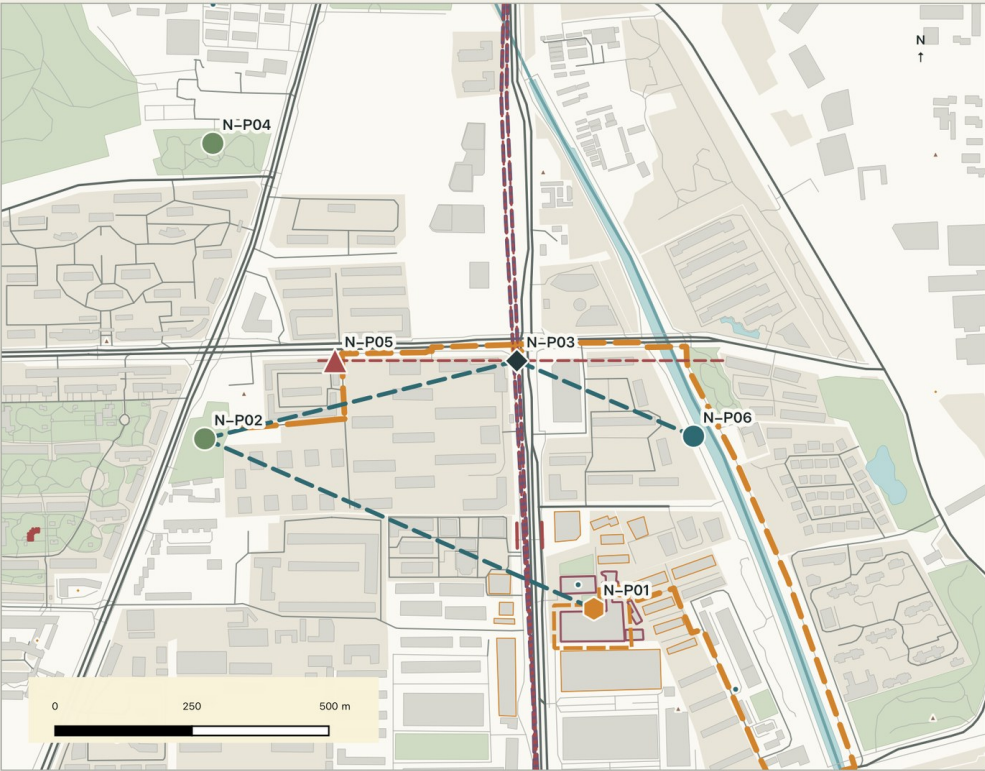
EXISTING / 公开映射现状

mapped ≠ verified



PROPOSED / 概念设计叠加

6 concepts - survey A5/B15



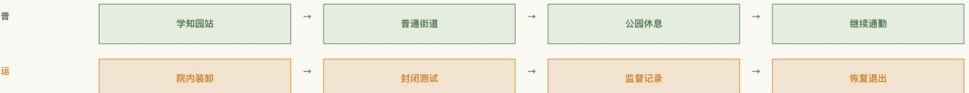
PROJECT REGISTER / 六个项目与关键门槛

项目	关键门槛
N-P01	机电研究所受控测试院
N-P02	逸成公园恢复与观察点
N-P03	学知园站普通到达点
N-P04	八家口袋公园安静休息点
N-P05	东西向断点审计门
N-P06	小月河水岸接口核查点

SECTION / 机制剖面 · 高程未测



TWO LANES / 普通人与测试运营分离



OSM orientation / EPSG:4548 / official locator ≠ parcel / test envelope = mapped building + 12m graphic buffer / demolish-retain UNASSIGNED

NOT SUBMISSION READY

Beijing AI Origin: Innovation Testbed

Build an Origin—heritage—street learning chain independent of campus access

03—M

AI原点 | 不依赖校园开放的原点—遗产—街道学习链

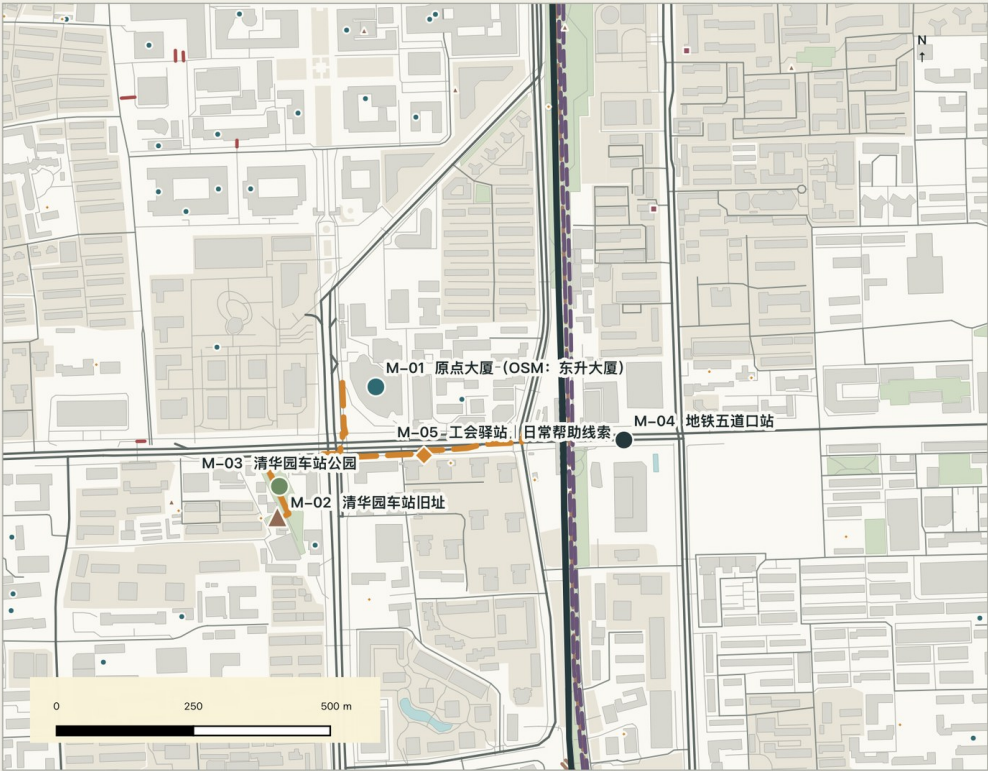
104.3 ha detailed boundary pending / mapped window 1.8 x 1.4 km / both maps 1:10000

A3 REVIEW V1

BOUNDARY PENDING / 不提交

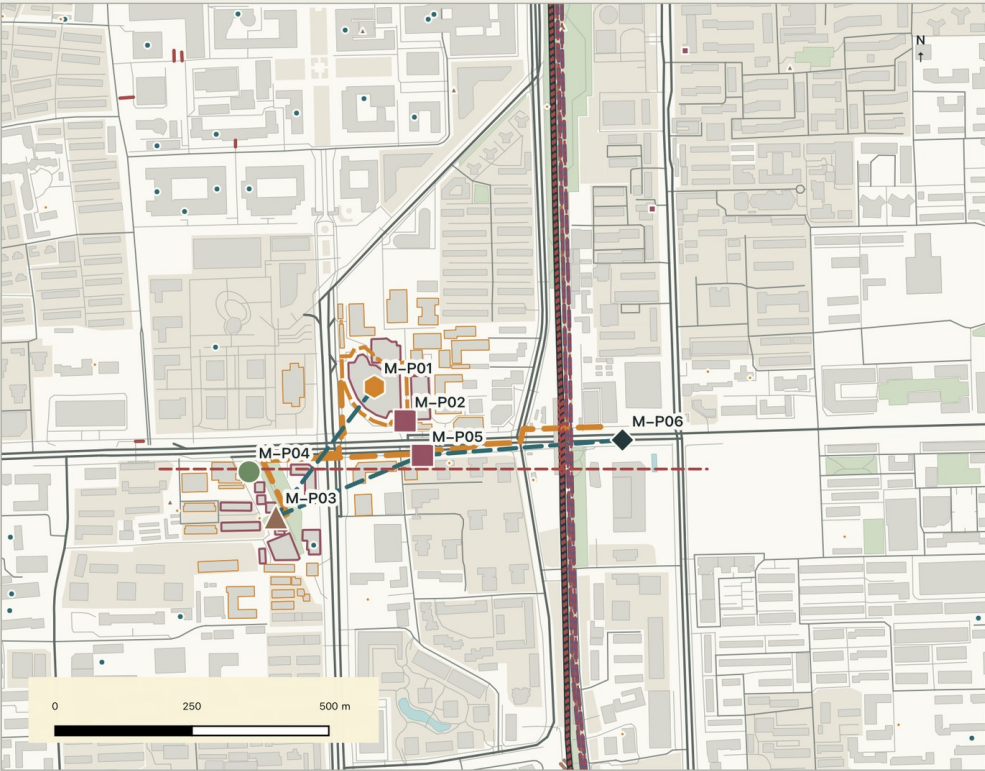
EXISTING / 公开映射现状

mapped ≠ verified



PROPOSED / 概念设计叠加

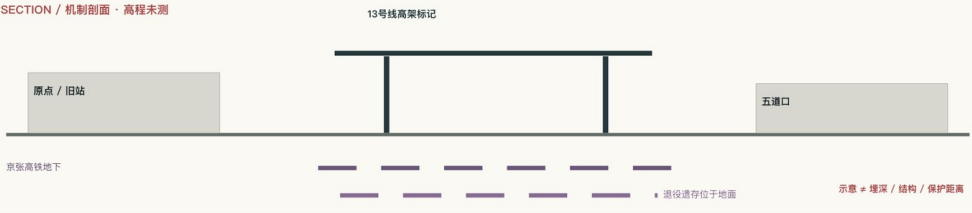
6 concepts - survey A15/B37



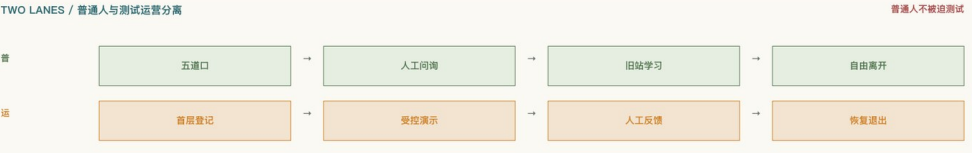
PROJECT REGISTER / 六个项目与关键门槛

	项目	关键门槛
M-P01	原点大厦受控首层试验室	楼宇许可、消防、电力
M-P02	原点人工问询台	首层开放、值守、语言
M-P03	清华园旧站无屏学习廊	文保边界、开放、展陈许可
M-P04	车站公园小组学习台	树木、园路、照明
M-P05	工会驿站日常帮助点	现有服务核实、开放、值守
M-P06	五道口站校—园—街门槛	站口客流、过街、校园门禁

SECTION / 机制剖面 · 高程未测



TWO LANES / 普通人与测试运营分离



OSM orientation / EPSG:4548 / official locator ≠ parcel / test envelope = mapped building + 12m graphic buffer / demolish-retain UNASSIGNED

NOT SUBMISSION READY

Dazhongsi: Urban Life Testbed

A no-phone station-to-service journey; non-diagnostic and human-accountable

03-S

大钟寺 | 从四象限换乘到人工服务的无手机日常链

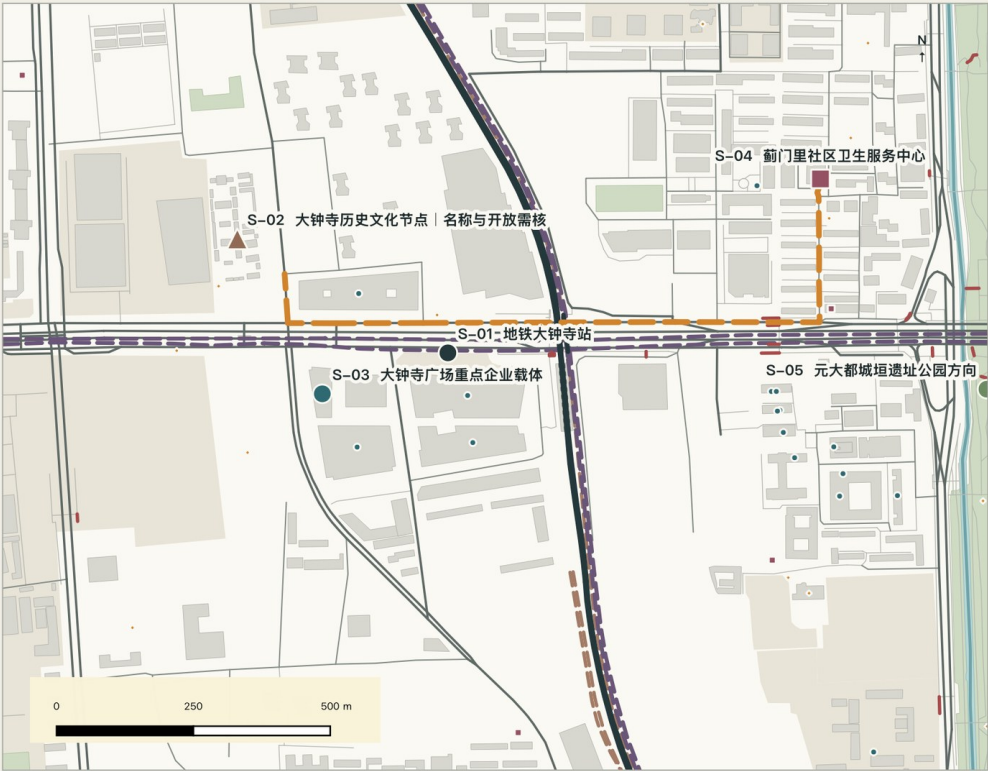
72.0 ha detailed boundary pending / mapped window 1.8 x 1.4 km / both maps 1:10000

A3 REVIEW V1

BOUNDARY PENDING / 不提交

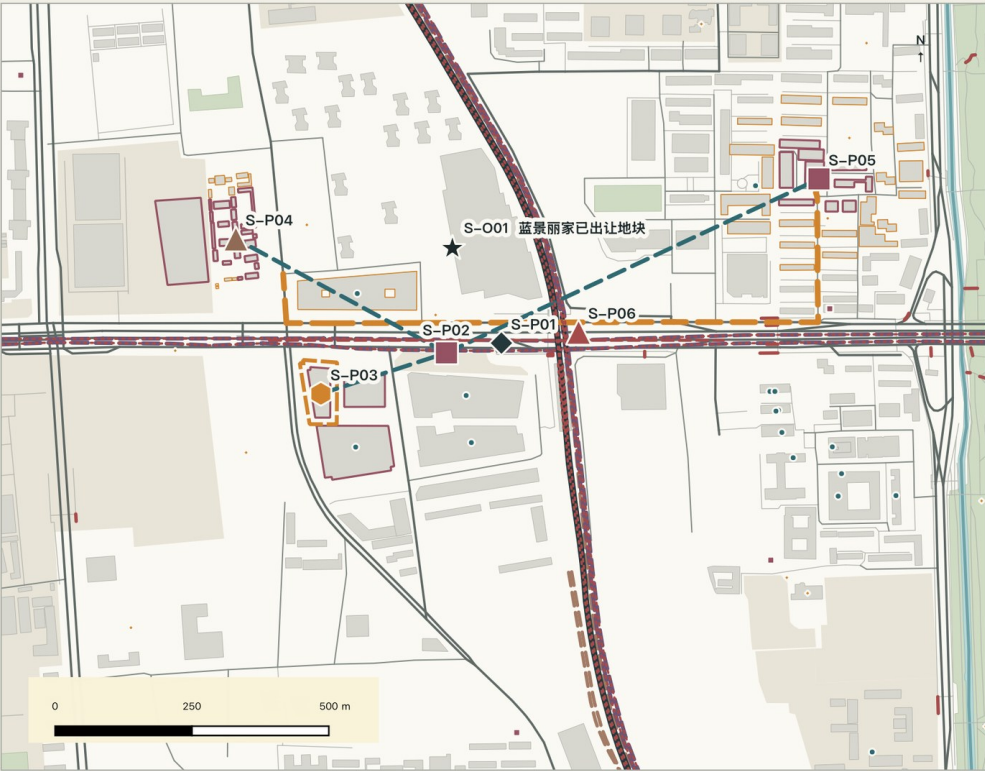
EXISTING / 公开映射现状

mapped ≠ verified



PROPOSED / 概念设计叠加

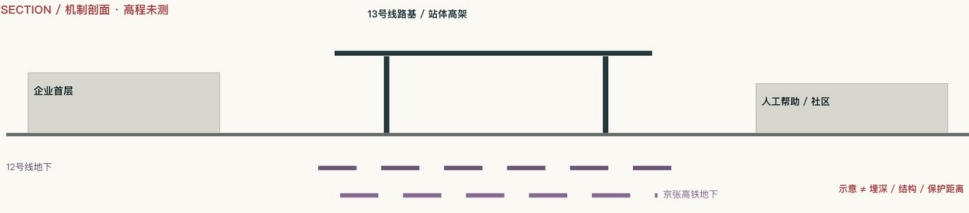
6 concepts + 1 official locator · A33/B32



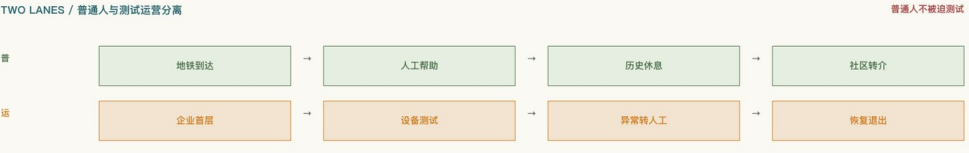
PROJECT REGISTER / 六个项目与关键门槛

项目	关键门槛
S-P01 大钟寺站四象限慢行链	客流、信号、桥柱/管线
S-P02 站口人工帮助窗	值守、运营审批、无障碍
S-P03 重点企业受控首层试实验室	业主许可、消防、装卸
S-P04 大钟寺历史休息与饮水点	名称/开放、文保、容量
S-P05 社区健康人工转介点	机构同意、医疗边界、隐私
S-P06 站口路缘与自行车管理点	分时观察、道路红线、执法/运营

SECTION / 机制剖面 · 高程未测



TWO LANES / 普通人与测试运营分离



OSM orientation / EPSG:4548 / official locator ≠ parcel / test envelope = mapped building + 12m graphic buffer / demolish-retain UNASSIGNED

NOT SUBMISSION READY

Mobility, rail verticality, and blue-green public space

Where no lawful crossing exists, the concept line stops at the operating boundary

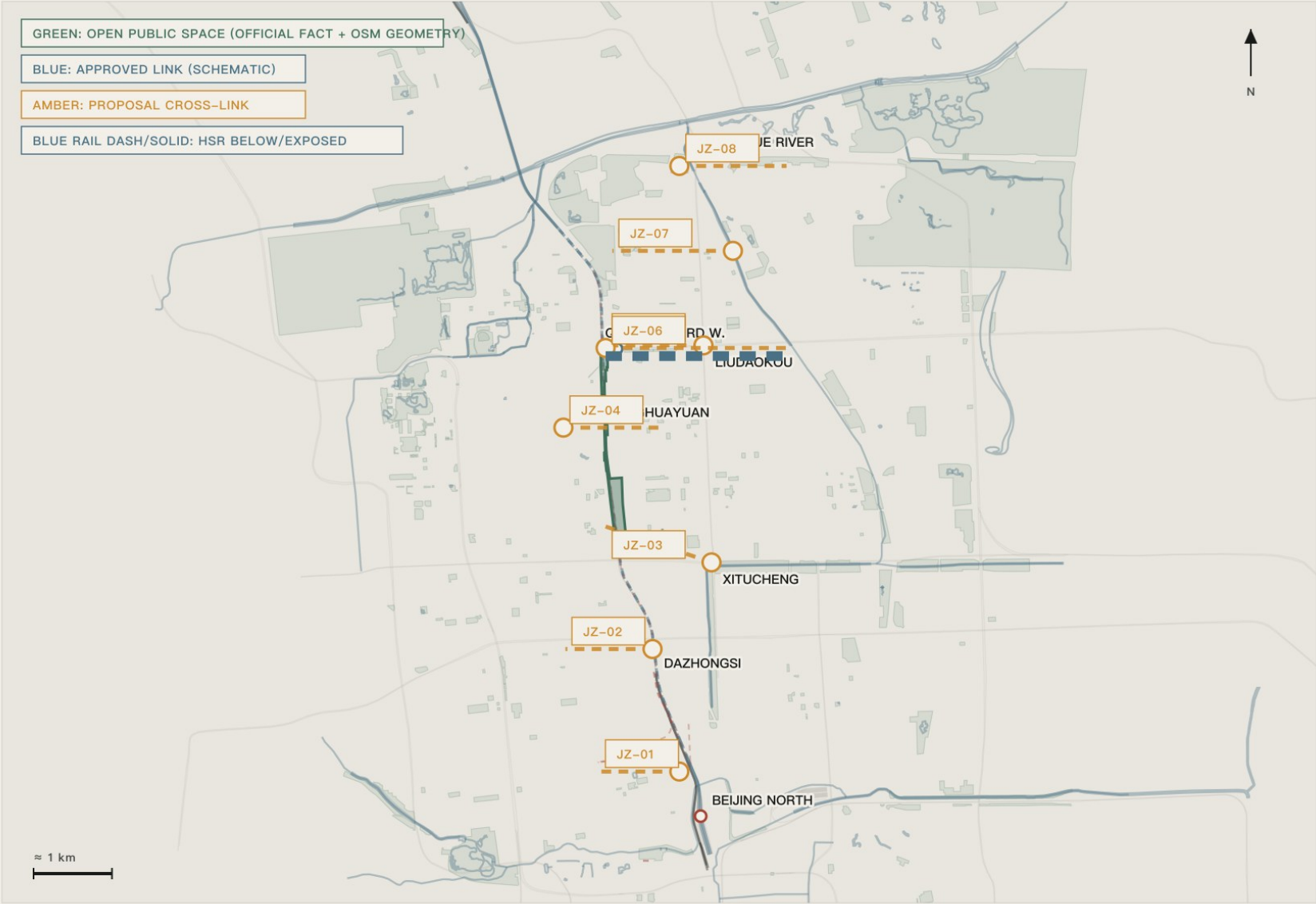
JZ-RCT / 04 / V0.1

04 / REAL PUBLIC SPINE × CROSS-BLUE-GREEN NETWORK

STATUS-AWARE NETWORK

NOT ONE GLOWING AI LINE A STATUS-AWARE PUBLIC NETWORK

Distinguish open, approved and merely mapped segments.
Then use eight cross-links to organise
station–park–campus–street–service journeys.



FIVE COMPLETE JOURNEYS

- 01 **STATION → PARK → SERVICE**
STATIC WAYFINDING / REST / CROSSING / HUMAN HELP
- 02 **CAMPUS → PUBLIC EDGE → COMMUNITY**
RESEARCH ACCESS IS NOT HIDDEN BEHIND CAMPUS ID
- 03 **ORDINARY PATH // BOUNDED TRIAL**
REFUSAL DOES NOT COST ACCESS OR SERVICE
- 04 **JING-ZHANG → COLLEGE RD → XIAOYUE**
APPROVED LINK AND PROPOSAL TRIAL STAY DISTINCT
- 05 **SERVICE → RETURN**
NO CONTINUOUS TRACKING; ONE-TIME, CANCELLABLE HELP

DRAWING DISCIPLINE

NO CONTINUOUS OPEN EVIDENCE → NO CONTINUOUS GREEN SPINE

NO FIELD CHECK → NO ACCESSIBILITY COVERAGE RATE

NO PERMIT → NO ROBOT IN PUBLIC RIGHT-OF-WAY

Basemap and park geometry: OpenStreetMap contributors, 2026–08–10. Open/approved status: official Beijing and Haidian sources. Cross-links are concepts, not road or engineering alignments.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL

Flagship 01: AI Learning Corridor

Problem Desk, no-screen explanation, Reverse Lab, Failure and Revision Library

JZ-RCT / SCENE-LEARNING / V0.1

FLAGSHIP 01 / AI LEARNING CORRIDOR

THE CITY ASKS FIRST. THE LAB ANSWERS SECOND.

A human hosts the problem desk; AI only offers optional grouping and explanation.
People can submit, understand and reach a human without a login or QR code.

AI-GENERATED CONCEPT SCENE / NOT A SITE PHOTO / NOT AN APPROVAL



SPATIAL RELATION

- 1 CONTINUOUS ORDINARY PATH + SHADE
- 2 HUMAN-HOSTED PROBLEM DESK
- 3 OPTIONAL, SWITCHABLE AI TERMINAL
- 4 RAIL HERITAGE + PUBLIC LEARNING SPACE

REAL TASK

Let people confirm the real urban problem instead of allowing a technical team to rewrite it for them.

STOP NOW

Stop if a request is distorted, identity leaks, urgency is downgraded, or no human owns the response.

WHAT REMAINS WITHOUT AI

Paper problem desk, places to sit, no-screen explanation and a public failure/revision archive.

NOW: V0.1 / NO FIELD TRIAL

Concept image explains spatial and role relationships only. It is not existing condition, site permission, partnership, device selection or final design.
Generated with built-in OpenAI image generation. Exact copy, data and boundaries remain deterministic vector overlays.

LOCAL CANDIDATE / DO NOT SUBMIT

Flagship 02: Embodied AI Urban Lab

Enclosed and slow; people first; physical stop; human takeover; restoration

JZ-RCT / SCENE-EMBODIED / V0.1

FLAGSHIP 02 / EMBODIED AI URBAN LAB

LEARN TO YIELD. LEARN TO FAIL SAFELY.

An ordinary path runs beside a bounded test cell.
On-site people take over every failure; passers-by are not test conditions.

AI-GENERATED CONCEPT SCENE / NOT A SITE PHOTO / NOT AN APPROVAL



Concept image explains spatial and role relationships only. It is not existing condition, site permission, partnership, device selection or final design.
Generated with built-in OpenAI image generation. Exact copy, data and boundaries remain deterministic vector overlays.

SPATIAL RELATION

- 1 ORDINARY PATH NEVER OCCUPIED
- 2 LOW-SPEED BOUNDED TEST CELL
- 3 ON-SITE MARSHAL + PHYSICAL STOP
- 4 RECOVERY BAY OFF THE PUBLIC PATH

REAL TASK

Test whether a device yields, degrades, stops and can be taken over — not merely whether it can move.

STOP NOW

Stop on any injury/near miss, ordinary-path incursion, failed emergency stop or retained identity data.

WHAT REMAINS WITHOUT AI

Clearer right-of-way, handoff and recovery rules, plus public near-miss and takeover records.

NOW: V0.1 / NO FIELD TRIAL

LOCAL CANDIDATE / DO NOT SUBMIT

Flagship 03: AI Health and Aging Lab

No-phone service finding, rest-to-service routes, caregiver human handoff

JZ-RCT / SCENE-HEALTH / V0.1
FLAGSHIP 03 / AI HEALTH & AGING LAB

COMPLETE THE JOURNEY WITHOUT A SMARTPHONE.

Seat, shade, water, static cues and a staffed window come first.
AI explains verified services only: no diagnosis, score or continuous tracking.

AI-GENERATED CONCEPT SCENE / NOT A SITE PHOTO / NOT AN APPROVAL



SPATIAL RELATION

- 1 SEAT, WATER + STEP-FREE PATH
- 2 STATIC DIRECTION, NO PHONE
- 3 ACCOUNTABLE HUMAN SERVICE WINDOW
- 4 OPTIONAL ONE-SESSION VOICE KIOSK

REAL TASK

Complete finding, reaching, resting, human handoff and return — not simply add a smart interaction.

STOP NOW

Stop on high-risk misdirection, unnecessary identity/medical data, unavailable human help or continuous location.

WHAT REMAINS WITHOUT AI

Verified paper directory, a rested route, static wayfinding, human help and a clear return path.

NOW: V0.1 / NO FIELD TRIAL

Concept image explains spatial and role relationships only. It is not existing condition, site permission, partnership, device selection or final design.
Generated with built-in OpenAI image generation. Exact copy, data and boundaries remain deterministic vector overlays.

LOCAL CANDIDATE / DO NOT SUBMIT

Twelve scenarios × City Release

Every card begins with ordinary baseline, permission, stop, exit, and public return

SC01

City Problem Desk

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC02

No-Screen Explanation Walk

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC03

Reverse Lab

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC04

Failure and Revision Library

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC05

Shared Right-of-Way Sandbox

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC06

Curb Handoff Bay

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC07

Failure and Takeover Drill

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC08

Non-Identifying Public-Realm Check

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC09

No-Phone Service Finder

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC10

Rest-to-Service Route

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC11

Caregiver Human Handoff

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

SC12

Service Directory Audit

Retain a better ordinary route, verified directory, maintenance record, or reusable failure evidence after exit.

Metrics, evidence, unknowns, and next actions

Recomputable does not mean official; FAR, height, density, and field performance remain unknown

JZ-RCT / 05 / V0.1

05 / INNOVATION × DAILY LIFE: EVIDENCE DECIDES WHAT COMES FIRST

RELATIVE PROXY / NOT OFFICIAL

157 HIGH-I × LOW-LIFE CELLS
MEAN FIELD CHECKS, NOT AI PROJECTS

These are field-verification candidates
under current public mapped proxies.
Ordinary-city repair first; conditional AI second.



EVIDENCE LEDGER

906

STUDY-WINDOW CELLS

≈200–250 m samples; not parcels

587 / 38 / 281

HIGH / MED / LOW CONFIDENCE

Low confidence is greyed in formal figures

98

INNOVATION NETWORK NODES

176 internal/semantic items excluded

40

OFFICIAL NEAR-CORRIDOR ELDER STATIONS

Unverified coordinates; excluded from score

DIAGNOSIS → SPATIAL ACTION

HIGH-I × LOW-LIFE: VERIFY ORDINARY SERVICES; DO NOT AUTO-DEPLOY AI;
LOW-I × LOW-LIFE: FILL EVIDENCE GAPS; UNKNOWN IS NOT ZERO;
K1 conflicts with its label; K2 varies internally; K3 geometry drifts;
EVERY TESTBED NEEDS FIELD, PERMISSION, OPERATOR AND EXIT RESOURCE.

MAIN: S0 / 800 M / DL4

SENSITIVITY: S1 / 400–1200 M / DL5

Data: OpenStreetMap contributors, common baseline 2026-08-10T03:51:58Z; official Haidian elder-service lists. Median-relative quadrants are not official service, capacity, quality or need assessments.

BILINGUAL REVIEW PACKAGE / CONCEPT—NOT AN APPROVAL